Math Exam 10 August, 2026

Total Marks: 20 **Time: 40 minutes**

[**Instructions**: Solve each problem clearly and show your solution step by step. Clearly write all necessary calculations and supporting work.]

1. What is the greatest number that always divides the product of three consecutive natural numbers?

2. What is the simple fraction of $45.2\dot{3}4\dot{6}$?

3. A tank has three pipes. The first and second pipes can fill the tank in 30 minutes and 20 minutes, respectively. The third pipe can empty the full tank in 60 minutes. If all three pipes are opened together, in how many minutes will the tank be filled?

4. Robin covers 480 km in 12 days by walking 10 hours daily. In how many days can he cover 360 km by walking 9 hours daily?

5. A can complete a piece of work in 30 days and B can complete it in 20 days. After half of the work was completed by both working together, B left. After half of the remaining work was finished by A working alone, B returned and they completed the remaining work together. How many total days were required to complete the work?

6. What is the square root of 26.5225?

7. What is the minimum number of soldiers that must be removed from a group of 56728 soldiers so that the troops can be arranged in a square?

8. Which smallest number, when divided by 20, 25, 30, 36, and 48, leaves remainders of 15, 20, 25, 31, and 43 respectively?

9. What is the greatest number that divides 100 and 184 leaving a remainder of 4 in each case?

10. In a school, the ratio of the number of boys and girls is 5 : 7. If the number of girls in that school is 350, what is the number of boys?

11. 180 mangoes were bought from a fruit shop. After two days, 9 mangoes rotted. What percentage of mangoes are good?

12. A fish seller bought Hilsa fish at 1600 Taka per hali (four pieces) and sold each fish for 350 Taka. What was his percentage of profit or loss?

13. A tea trader buys a box of tea leaves at the rate of 80 Taka per kg. Selling all the tea leaves at the rate of 75 Taka per kg results in a loss of 500 Taka. How many kilograms of tea leaves did he buy?

14. Determine the compound interest on 5000 Taka for 2 years at an annual interest rate of 10.50%.

15. At a simple interest rate of 8.5% per annum, what amount of principal will yield an interest of 2550 Taka in 6 years?

16. The average age of Tanvir's family consisting of 5 members 3 years ago was 35 years. One year ago, a new baby was born in this family. What will be the average age of the family three years hence?

17. The average salary of all the 60 employees in an office is 12,000 taka per month. If the number of executives is twice the number of non-executive employees and if the average salary of non-executives be 2/5th of the average salary of executives, then what is the average salary of non-executive employees?

18. Samia travels the first one-third of a journey at a speed of 10 km/hr, the next one-third at a speed of 20 km/hr, and the final one-third at a speed of 60 km/hr. What is Samia's average speed for the entire journey?

19. Ratul travels 30 minutes at the speed of 25 km/hr. Further he travels 20 minutes at the speed of 40 km/hr. Find his average speed.

20. Modern electronic shop sold 30% hardware at the profit of 50% and 90% software at the profit of 10%. What is the average profit per cent of the Modern electronic shop is, if it sells only these two kinds of things?